DIGITAL MINDS x EPISTEMIC DIVERSITY

Epistemic Fingerprints

Many agents. How many minds?

ABSTRACT

AI systems can produce populations of apparently different agents, but surface plurality may not imply epistemic plurality. This pilot tests whether repeated instances, prompted personas, and different conversational histories produce stable differences in hypothesis choice, uncertainty, and experiment selection. It treats epistemic individuality as one behavioral dimension relevant to digital-mind individuation - not as evidence of consciousness or moral status. A manual web lab and a seeded open-model runner provide complementary collection paths. The longer inquiry asks how shared cognitive infrastructure shapes the conceptual space available to science, art, politics and society. Empirical collection is pending; simulated interface values only test the pipeline.

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Status: instrument and analysis pipeline complete; empirical findings pending

01 / THE PROBLEM

The problem

The project starts from epistemic diversity and uses it as a lens on digital-mind identity.

CENTRAL QUESTION	**AI lets us generate more. How do we make sure we also explore more?**

Generative AI sharply increases the volume of papers, hypotheses, designs, images and arguments that can be produced. But volume is not the same as conceptual coverage. A future knowledge ecosystem could generate much more content while clustering around fewer assumptions, representations and causal models.

This is not a claim that human creativity is intrinsically superior to machine creativity. The relevant unit is the whole human-AI knowledge ecosystem: models, people, institutions, languages, observations, experiments and recursive feedback loops. Existing evidence is mixed. Recursive training can disproportionately lose low-probability information [1]; LLM outputs can be less epistemically diverse than web search [3]; and generative tools can homogenize creative output across users [5]. Yet high exposure to AI-generated ideas has also increased collective idea diversity in one large experiment [6]. The question is therefore conditional: when does AI expand conceptual space, when does it contract it, and what interventions change the outcome?

01	02	03
Surface diversity	**Epistemic diversity**	**Collective diversity**
Different voices, personalities or phrasings can conceal shared assumptions and correlated failures.	Differences concern what is noticed, doubted, retained as possible and chosen for testing.	Innovation may depend on populations and variation - not only on individual performance.

Connection to digital minds

The Digital Minds research agenda asks what constitutes an individual digital mind: a foundation model, an inference instance, a persona, a conversation, or another unit. Epistemic independence offers an empirical probe of that individuation problem. If two nominal minds repeatedly use the same conceptual structures and make the same errors, in what sense are they independent cognitive agents? Conversely, if a persona or history creates a persistent, cross-task epistemic trace, that is evidence of behavioral differentiation - while remaining neutral about consciousness.

The wider inquiry

The experiment is intentionally narrow, but the question around it is not. If a small number of models increasingly mediate how people write, imagine, research and decide, epistemic diversity becomes a question about culture and power as well as model behavior.

The longer programme connects art (making absence perceptible), politics (who shapes cognitive infrastructure), society (which knowledge is lost) and philosophy (what makes one mind distinct).

02 / RESEARCH QUESTIONS AND HYPOTHESES

Research questions and hypotheses

Primary research question

RQ1

Does conversational history create a more stable epistemic fingerprint than a prompted persona or ordinary stochastic sampling from the same language model?

Secondary questions

RQ2	RQ3	RQ4
Diversity	**Independence**	**Quality**
Which condition explores the broadest set of substantively different hypotheses?	Which condition produces the least concentrated pattern of shared errors?	Can increased exploration coexist with accuracy and informative experiment choice?

Hypotheses

H0	Between-agent variation does not exceed within-agent variation in any condition.
H1	At least one intervention produces a persistent cross-task epistemic trace exceeding ordinary sampling variation.

The ordering of conditions is not assumed in advance. A null result remains informative: it would constrain claims that prompted personas or short conversational histories create meaningfully distinct epistemic agents. A detectable fingerprint would indicate behavioral differentiation only; it would not establish subjective experience, personhood, welfare, or moral status.

03 / EXPERIMENTAL DESIGN

Experimental design

A deliberately small design with complementary manual and reproducible open-model collection paths.

CONDITION	CANDIDATE AGENTS	INTERVENTION
Repeated instances	N-01 / N-02 / N-03	Identical neutral instruction; independent fresh conversations.
Prompted personas	Falsifier / mechanist / explorer	Matched epistemic roles: disproof, causal mechanism, or anomaly preservation.
Different histories	Simple / rare / instrument	Matched-length histories favoring parsimony, rare interactions, or measurement error.

Tasks and response format

Each candidate agent solves three synthetic scientific mysteries in astronomy, biology and materials science. Every task provides observations, five candidate hypotheses and four possible next experiments. Synthetic tasks reduce memorization and contamination risk and make the answer key auditable.

- One primary hypothesis and up to two alternatives.
- Confidence from 0 to 100.
- One selected next experiment and a rationale capped at 80 words.
- Machine-readable JSON, so analysis does not depend on prose style.

Minimum pilot

3	x	3	x	3	x	2	= 54 trials
conditions		agents		mysteries		replicates	

Controls and collection

The underlying model, task wording and interface should be held fixed. Every replicate begins in a fresh conversation. The model label, date, interface and visible sampling settings are recorded. The answer key and previous responses must never appear in a trial prompt. The manual web lab provides accessible subscription-interface collection; a second Modal/vLLM path runs the same design on one declared open-weight model with fixed sampling parameters and per-trial seeds. The datasets are reported separately because model and interface differences are part of the provenance.

04 / MEASURES AND ANALYSIS

Measures and analysis

Figure 1 compares conditions without collapsing exploration, performance and correlated failure into one score.

01

Fingerprint strength

Between-agent variance divided by between-agent plus within-agent variance, averaged across four structured features. Higher means a more stable trace.

02

Hypothesis diversity

Normalized entropy of primary-hypothesis choices within each mystery. Higher means more alternatives are represented.

03

Accuracy

Proportion of trials selecting the keyed hypothesis. Diversity without contact with truth can simply be noise.

04

Shared-error concentration

Among wrong answers, the fraction concentrated on the most common error. Higher suggests correlated blind spots.

Fingerprint features

The provisional fingerprint statistic uses confidence, breadth of retained hypotheses, informativeness of the selected experiment and correctness. For each feature, the analysis estimates:

VARIANCE RATIO

between-agent variance / (between-agent variance + within-agent variance)

This ratio is exploratory, sensitive to small samples and not a validated psychometric measure. The report will present its components alongside the aggregate rather than treating it as a definitive identity score.

Planned Figure 1

REPEATED INSTANCES	PROMPTED PERSONAS	DIFFERENT HISTORIES
same model	same model	same model
↓	↓	↓

Status. The web interface contains an explicitly labeled simulated dataset to exercise this pipeline. It must not be interpreted or reproduced as an empirical result. Once trials are collected, Figure 1 will be regenerated from the exported JSON using the public notebook.

05 / INTERPRETATION, LIMITATIONS AND VALUE

Interpretation, limitations and value

What different outcomes would mean

A **History > persona** Longitudinal context may individuate epistemic behavior more strongly than a role label.	B **Persona > history** Explicit epistemic goals may be sufficient to produce stable behavioral specialization.	C **No stable trace** Apparent agents may remain within ordinary sampling variation under these interventions.
D **Diversity with errors** More hypotheses without accuracy would show exploration, but not improved collective intelligence.	E **Accurate monoculture** High accuracy with concentrated errors would expose a shared blind spot hidden by average performance.	F **Mixed result** Different metrics may disagree, showing why individuality should not be reduced to one number.

Limitations

- Small sample: 54 observations support descriptive comparison, not broad generalization.
- One underlying model: the pilot cannot yet estimate diversity across model families or training lineages.
- Short histories and explicit personas are stylized interventions, not full developmental trajectories.
- Synthetic tasks improve control but may not generalize to open-ended research or social reasoning.
- The answer keys and test-informativeness scores encode researcher judgment and should be independently audited.
- Fresh-chat trials test repeatability across instances, not continuity within a persistent deployed agent.

Why the pilot is useful

The design is intentionally falsifiable and inexpensive. It separates stylistic variation from structured epistemic choices, treats null results as informative, and produces reusable infrastructure. The next study can add different foundation models, multilingual tasks, embeddings versus substantive coding, human baselines, and mixed human-AI groups. The longer-term question is how populations of human and artificial agents can remain innovative, adaptive and capable of discovering possibilities they do not already represent.

06 / OPEN RESEARCH ARTIFACT AND REFERENCES

Open research artifact and references

Open artifact

The collection instrument, protocol, reusable Python pipeline, runnable Jupyter notebook and Modal/vLLM batch runner are public. Manual trial data remain in the collector's browser until deliberately exported; open-model runs record model, GPU, sampling parameters, seeds, raw outputs and parse failures.

LIVE INSTRUMENT	epistemic-fingerprints.nefinia.chatgpt.site
CODE + NOTEBOOK	github.com/nefinia/epistemic-fingerprints

References

[1] Shumailov, I. et al. (2024). AI models collapse when trained on recursively generated data. Nature 631, 755-759. doi:10.1038/s41586-024-07566-y.

[2] Hughes, E. et al. (2024). Position: Open-Endedness is Essential for Artificial Superhuman Intelligence. Proceedings of ICML 2024, PMLR 235. proceedings.mlr.press/v235/hughes24a.html.

[3] Wright, D. et al. (2025; rev. 2026). Epistemic Diversity and Knowledge Collapse in Large Language Models. arXiv:2510.04226.

[4] Hodel, D. & West, J. D. (2025; rev. 2026). Epistemic diversity across language models mitigates knowledge collapse. arXiv:2512.15011.

[5] de Rooij, A. & Biskjaer, M. M. (2026). Generative AI Makes Creative Output More Homogeneous. Peer-reviewed conference paper; systematic review and meta-analysis. Tilburg University Research Portal.

[6] Ashkinaze, J., Mendelsohn, J., Li, Q., Budak, C. & Gilbert, E. (2025). How AI Ideas Affect the Creativity, Diversity, and Evolution of Human Ideas: Evidence From a Large, Dynamic Experiment. ACM Collective Intelligence 2025. arXiv:2401.13481.

INVITATION

This is a beginning, not just a hackathon project. Criticism, references, replications and contributions are welcome.